Delivering Military Advantage through multi-national geospatial interoperability

DGIWG-NWP-LIGHTNING-001

Edition 1.0

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DGIWG NWP Earth Observations Lightning Profile

CONTENTS

I. ABSTRACT

This document defines the DGIWG NWP Earth Observations Lightning Profile, an OGC API — Environmental Data Retrieval (EDR) Part 3 Service Profile. It specifies normative requirements and conformance tests for server implementations conforming to this profile.

II. KEYWORDS

The following are keywords to be used by search engines and document catalogues.

cube, radius, items, datetime, frequency_of_lightning_flashes_per_unit_area, lightning flashes, lightning

III. PREFACE

This document was prepared by Met Office, DGIWG.

IV. SECURITY CONSIDERATIONS

No security considerations have been made for this document.

V. SUBMITTING ORGANIZATIONS

The following organizations submitted this Document to the Open Geospatial Consortium (OGC):

- Met Office
- DGIWG

VI. SUBMITTERS

All questions regarding this document should be directed to the editor or the submitters:

Table — Submitters

NAME	AFFILIATION
Austria	Institute of Military Geography
France	Institut National de l'Information Géographique et Forestière (IGN)
Finland	Finnish Defence Forces
Germany	Bundeswehr Geoinformation Centre (BGIC)
NATO	NATO Maritime GEOMETOC Centre of Excellence
United Kingdom	Met Office (MO), UK Strategic Command, DSTL
United States	National Geospatial-Intelligence Agency (NGA)

1. SCOPE

This standard defines the DGIWG NWP Earth Observations Lightning Profile. It specifies requirements and conformance tests for implementations of this profile.

2. CONFORMANCE

Conformance with this standard shall be checked using the Abstract Test Suite in Annex A.

The following conformance classes are defined:

- https://schemas.dgiwg.org/edr/1.0/conf/nwp_earth_observations_lightning

3. NORMATIVE REFERENCES

The following documents are referred to in the text in such a way that some or all of their content constitutes requirements of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

Mark Burgoyne, David Blodgett, Charles Heazel, Chris Little, Open Geospatial Consortium: OGC 19-086r6, *OGC API — Environmental Data Retrieval Standard*. Open Geospatial Consortium (2023). http://www.opengis.net/doc/IS/ogcapi-edr-1/1.1.0.

OGC API — EDR Part 3: Service Profiles (draft)

4. TERMS AND DEFINITIONS

No terms and definitions are listed in this document.

This document uses the terms defined in OGC Policy Directive 49, which is based on the ISO/IEC Directives, Part 2, Rules for the structure and drafting of International Standards. In particular, the word "shall" (not "must") is the verb form used to indicate a requirement to be strictly followed to conform to this standard.

This document also uses terms defined in OGC API — EDR Part 1: Core.

5. REQUIREMENTS

REQUIREMENTS CLASS 1	
IDENTIFIER	`https://schemas.dgiwg.org/edr/1.0/req/nwp_earth_observations_lightning`
CONFORMANCE CLASS	Conformance class A.1: `https://schemas.dgiwg.org/edr/1.0/conf/nwp_earth_observations_lightning`
TARGET-TYPE	DGIWG NWP Earth Observations Lightning Profile Profile Standard
NORMATIVE STATEMENTS	Requirement 1: `/req/nwp_earth_observations_lightning/items-endpoint` Requirement 2: `/req/nwp_earth_observations_lightning/cube-endpoint` Requirement 3: `/req/nwp_earth_observations_lightning/radius-endpoint`

REQUIREMENT 1	
IDENTIFIER	`/req/nwp_earth_observations_lightning/items-endpoint`
INCLUDED IN	Requirements class 1: `https://schemas.dgiwg.org/edr/1.0/req/nwp_earth_observations_lightning`
STATEMENT	The service SHALL provide a /collections/earth_observations_lightning/items endpoint.
A	The service SHALL return GeoJSON FeatureCollection.
B	Each feature SHALL include frequency_of_lightning_flashes_per_unit_area property.
C	Each feature SHALL include datetime property in the format YYYY-MM-DDTHH:MM:SSZ e.g. 2026-04-12T21:20:50.52Z.

REQUIREMENT 2	
IDENTIFIER	`/req/nwp_earth_observations_lightning/cube-endpoint`
INCLUDED IN	Requirements class 1: `https://schemas.dgiwg.org/edr/1.0/req/nwp_earth_observations_lightning`
STATEMENT	The service SHALL provide a /collections/earth_observations_lightning/cube endpoint.
A	The service SHALL return CoverageJSON FeatureCollection.

REQUIREMENT 2	
B	Each feature SHALL include frequency_of_lightning_flashes_per_unit_area property.
C	Each feature SHALL include datetime property in the format YYYY-MM-DDTHH:MM:SSZ e.g. 2026-04-12T21:20:50.52Z.

REQUIREMENT 3	
IDENTIFIER	`/req/nwp_earth_observations_lightning/radius-endpoint`
INCLUDED IN	Requirements class 1: `https://schemas.dgiwg.org/edr/1.0/req/nwp_earth_observations_lightning`
STATEMENT	The service SHALL provide a /collections/earth_observations_lightning/radius endpoint.
A	The service SHALL return CoverageJSON FeatureCollection.
B	Each feature SHALL include frequency_of_lightning_flashes_per_unit_area property.
C	Each feature SHALL include datetime property in the format YYYY-MM-DDTHH:MM:SSZ e.g. 2026-04-12T21:20:50.52Z.

ANNEX A
(NORMATIVE)
ABSTRACT TEST SUITE

CONFORMANCE CLASS A.1	
IDENTIFIER	`https://schemas.dgiwg.org/edr/1.0/conf/nwp_earth_observations_lightning`
REQUIREMENTS CLASS	Requirements class 1: `https://schemas.dgiwg.org/edr/1.0/req/nwp_earth_observations_lightning`
CONFORMANCE TESTS	Abstract test A.1: `/conf/nwp_earth_observations_lightning/items-endpoint` Abstract test A.2: `/conf/nwp_earth_observations_lightning/cube-endpoint` Abstract test A.3: `/conf/nwp_earth_observations_lightning/radius-endpoint`

ABSTRACT TEST A.1	
IDENTIFIER	`/conf/nwp_earth_observations_lightning/items-endpoint`
REQUIREMENT	Requirement 1: `/req/nwp_earth_observations_lightning/items-endpoint`
TEST PURPOSE	Validate that items endpoint is correctly implemented.
TEST METHOD	
STEP	Send GET request to /collections/earth_observations_lightning/items. Verify response Content-Type is application/json. Verify each feature contains frequency_of_lightning_flashes_per_unit_area property. Verify each feature contains datetime property.

ABSTRACT TEST A.2	
IDENTIFIER	`/conf/nwp_earth_observations_lightning/cube-endpoint`

ABSTRACT TEST A.2

REQUIREMENT	Requirement 2: `/req/nwp_earth_observations_lightning/cube-endpoint`
TEST PURPOSE	Validate that cube endpoint is correctly implemented.
TEST METHOD	
STEP	Send GET request to /collections/earth_observations_lightning/cube with a valid coordinate in crs_4326 and a radius of 1km. Verify response Content-Type is application/json. Verify each feature contains frequency_of_lightning_flashes_per_unit_area property. Verify each feature contains datetime property. Verify each feature is in projection crs_4326.

ABSTRACT TEST A.3

IDENTIFIER	`/conf/nwp_earth_observations_lightning/radius-endpoint`
REQUIREMENT	Requirement 3: `/req/nwp_earth_observations_lightning/radius-endpoint`
TEST PURPOSE	Validate that radius endpoint is correctly implemented.
TEST METHOD	
STEP	Send GET request to /collections/earth_observations_lightning/radius with a valid coordinate in crs_4326. Verify response Content-Type is application/json. Verify each feature contains frequency_of_lightning_flashes_per_unit_area property. Verify each feature contains datetime property. Verify each feature is in projection crs_4326. Verify each feature is returned within the radius area.